and **critical public health reporting requirements**. **GeneXpert and MGIT 960 integration is essential. Weekly growth monitoring interface must be user-friendly** for technicians logging results over 8 weeks.

VIROLOGY & VACCINE UNIT - DATA CONFIGURATION SUMMARY

LABORATORY OVERVIEW

Virology & Vaccine Unit performs viral diagnostics, vaccine development, and bacteriological testing related to vaccine production. Handles virus culture, vaccine production pathways, and quality control for biologics.

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1. SAMPLE TYPES

Incoming Samples:

- Clinical specimens (for viral testing)
- Viral isolates
- Cell culture samples
- Vaccine candidates
- Bacterial samples (for bacteriological testing related to vaccine production)

Generated Materials:

- Cultured viruses
- Vaccine products (various stages)
- Seed virus stocks
- Bacterial cultures and products

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2. KEY METADATA FIELDS

Sample Arrival:

- Sample ID
- Source (patient, animal model, environmental, production batch)

- Sample type
- Date/time of receipt
- Test type (viral vs. bacterial)
- Project/study association

Virus/Vaccine Production:

- Batch ID
- Production stage
- Cell line used
- Passage number
- Titer values
- Quality control results
- Formulation details

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3. COMPLETE WORKFLOW STAGES

VIRAL TESTING WORKFLOW

STAGE 1: Sample Intake & Registration

Sample Arrival:

- Pre-labeled or labeled upon arrival
- Can be clinical specimens or research samples

Registration:

- Metadata entry: Sample ID, Source, Type, Date/time
- Test type assignment: Viral testing
- Link to project/study if applicable

STAGE 2: Virus Culture Growth (In-Lab)

Process Steps:

Step	Description	LMIS Requirement

Media Preparation	Select media, reagents, equipment	Log materials used (type, lot number, expiry)
Sterilization	Autoclaving, filtration	Record sterilization parameters (temp, time, pressure)
Cell Culture	Grow host cells	Track cell line, passage number, growth conditions
Quality Control	Validate cell viability and sterility	Log QC results (viability %, sterility pass/fail)
Virus Culture	Inoculate cells with virus	Record virus strain, culture conditions (temp, CO_2, duration)
Dark Room Imaging	Imaging or fluorescence analysis	Store image data, analysis results (CPE observation, fluorescence intensity)
Formulation	Prepare viral product	Document formulation details (stabilizers, preservatives, concentrations)
Feeding	Maintain culture	Log feeding schedule, reagents used
Packaging	Final product packaging	Track batch ID, packaging specs (vial type, fill volume, labeling)

STAGE 3: Vaccine Development (Partially External)

Process Stages:

Stage	Description	LMIS Requirement
Virus Isolation	From culture	Link to culture batch ID
Titer Measurement	Quantify viral load	Record titer values (TCID50, PFU/ml, etc.)
Genome Sequencing	Viral genome analysis	Store sequence data (FASTA files, GenBank accession if submitted)
Seed Virus Production	Select strain for vaccine	Document selection criteria, seed virus batch ID
Preclinical Trials	Animal testing (external)	Track trial initiation date, animal species, outcomes (immunogenicity, safety)